CS MATH · UNIT L · THE CASINO LAB · QUARTER 3

The Casino Lab

Computer Science Math 2026–27 · in class **Jan 5–12** · demo **Thursday, January 14, 2027**

NAME _____ PARTNER _____ PERIOD _____

Driving question. The dice are fair and the deck is honest. People walk out ahead every night. So how does the casino still get paid — and how much does an evening cost *you*?
The plan. Play three games by hand for a few minutes, then build a Google Sheet that plays each one thousands of times. Your model, not your luck, answers the question.
Supplied: two dice, one deck, and 20 chips per pair. **You need:** a laptop with Google Sheets and the starter workbook. Chips have no cash value; this is a math model, not a gambling activity.

Learning goals

BY THE DEMO YOU CAN…

1. Turn a game's rules into a spreadsheet model: one row per round, helper columns, and a paytable the formulas look up.
2. Keep every assumption (payouts, bet, strategy) in a labelled parameter cell, so changing one cell re-prices the whole game.
3. Deal cards *without replacement*, and catch a model that silently produces impossible hands.
4. Compute expected value with SUMPRODUCT where the math allows it, and measure it by simulation where it does not.
5. Chart a running average converging on expected value, and turn a house edge into dollars per hour.

Outline and pacing

CHECKPOINTS, NOT SEPARATE GRADES

MEETING	IN CLASS	CHECKPOINT BEFORE YOU LEAVE
Tue Jan 5 72 min	Deck: Expected Value & the House Edge · 8 min in the simulator · **Casino floor:** play all three games, record every round, time yourselves	Part 1 recording tables full; minutes per game written down
Thu Jan 7 80 min	Import the starter · type in your physical rounds · **build Field Dice** to 10,000 rounds · start the card model and run the **bug hunt**	Dice EV + convergence chart; poker deal showing duplicates
Mon Jan 11 40 min	Fix the deal · **build Match Poker**: ranks, match score, paytable lookup, EV	Poker at 5,000 hands, 0 duplicates, frequencies vs exact
Tue Jan 12 80 min	**Build Blackjack** · try other strategies · Dashboard and dollars per hour · plan the demo	All three games on the Dashboard; write-up drafted
Thu Jan 14 80 min	**Casino Lab demo** (Part 8) · workbook link and write-up due at the end of class	

1. The casino floor

TUE JAN 5 · ABOUT 30 MIN

Setup. One partner is the **Player**, the other is the **House**; swap roles halfway through each game. The Player starts with **20 chips**; the House pays and collects from the bank. **Every game is Player vs House** — never partner vs partner — so every game has fixed rules and a real house edge. Record every round before the next one starts, and write down how many minutes each game took (Part 6 turns that into dollars per hour).

GAME 1 · 15 ROUNDS

Field Dice

Bet **1 chip**. Roll two dice once. The **sum** decides everything — each sum has exactly one result.

Sum 2	win +2
Sum 3, 4, 9, 10, 11	win +1
Sum 12	win +3
Sum 5, 6, 7, 8	lose −1

You win on 7 different sums and lose on only 4. Is that a good deal? Guess now; prove it Thursday.

GAME 2 · 10 ROUNDS

Blackjack vs the Dealer

Bet **2 chips**. **Shuffle the whole deck before every round.** Cards 2–10 count face value; J, Q, K count 10; an Ace counts 11 unless that would bust you, then 1.

1. Deal 2 to the Player, 2 to the Dealer (the House).
2. **Natural** (Ace + a 10-card): Player natural wins **+3**; Dealer natural wins for the House (−2); both = push.
3. Player **hits until 17 or more**. Over 21 = bust, lose −2 at once.
4. Dealer **hits until 17 or more**. Dealer bust = Player +2.
5. Otherwise higher total wins 2 chips; a tie is a push (0).

A hand stops at 5 cards. Both sides follow fixed rules — no choices — so a spreadsheet can play it exactly the same way.

GAME 3 · 10 ROUNDS

Match Poker

Bet **1 chip**. Shuffle, deal the Player 5 cards, read the paytable. No draws, no folding, no betting rounds. **Only matching ranks count** — straights and flushes are not on this table.

No match (5 different ranks)	lose −1
One pair	push 0
Two pair	win +5
Three of a kind	win +9
Full house (3 + 2)	win +20
Four of a kind	win +50

1 Record every round

NET = CHIPS THE PLAYER WON (+) OR LOST (−)

#	DIE 1	DIE 2	SUM	NET	#	DIE 1	DIE 2	SUM	NET
1					9				
2					10				
3					11				
4					12				
5					13				
6					14				
7					15				
8					DICE · MINUTES:				

#	PLAYER FINAL	DEALER FINAL	RESULT	NET
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
BLACKJACK · MINUTES:				

#	POKER HAND	NET
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
POKER · MINUTES:		

Before you build — predict. Rank the three games from best to worst for the Player, and say why in one line.

Which game did *you* finish ahead on? Does that make it a good bet?

2. Build Field Dice

THU JAN 7 · 35 MIN · TAB: DICE

Import the starter into Google Sheets (File → Import → Upload, or open it from Drive and *Save as Google Sheets*), then make your own copy. **Yellow** cells are yours to build; **blue** cells are parameters; **gray** is given. Type your Part 1 rounds into the *Physical Play* tab first — its formulas reuse each game's paytable, so your real rounds and your simulated rounds are scored by the same model.

2 The paytable does the thinking

LOOKUP · SUMPRODUCT

1. **Ways and probability.** In the paytable, the number of ways to roll a sum *s* is `6 − |s − 7|`. Build `J4` with that, then `K4 = J4/36`; fill down. The check cell must read 36.
2. **Exact expected value.** `L17`: multiply each probability by its payout and add — one `SUMPRODUCT`. House edge `L18 = −EV ÷ bet`.

3. **One round, row 2.** Die 1 and Die 2 are each `RANDBETWEEN(1,6)`; Sum adds them; Net looks the sum up in the paytable:

```
=VLOOKUP(D2, $I$4:$L$14, 4, FALSE)
```

The `$` signs pin the table so it does not slide when you fill down. No `IF` chains, no typed payouts.

4. **Running total and average.** `F2 = E2`, then `F3 = F2 + E3` filled down; `G2 = F2 / A2`. (Why not `SUM($E$2:E3)`? At row 10,000 that re-adds 10,000 cells, every row, every recalculation.)

5. **Scale.** Test at 20 rows. Fill to row 10,001 (10,000 rounds). Fill the results block.

6. **Chart.** Line chart of the running average (column G) vs round. Add the exact EV as a flat reference line (a helper column equal to `$L$17` works).

`ways total = 36` `exact EV = −1/36 ≈ −0.0278 chips` `P(player wins a round) = 44.4%`

`10,000-round average: usually −0.051 to −0.005`

Tune it. Change exactly one blue payout so the game becomes perfectly fair (EV = 0). Which cell, and to what? Then put it back.

3. Cards as numbers — and the bug hunt

THU JAN 7 · 30 MIN · TABS: DECK, POKER

A card is a number from 1 to 52. `Rank = MOD(card−1, 13) + 1` gives 1 (Ace) through 13 (King); `Suit = INT((card−1)/13) + 1` gives 1–4. The *Deck* tab lists all 52 so you can check yourself.

The bug hunt. Build the poker deal the "obvious" way first: `=RANDBETWEEN(1,52)` in each of the five card cells B2:F2. Then add the duplicate check in Q2:

```
=IF(SUMPRODUCT(COUNTIF(B2:F2,B2:F2))>5, "DUPLICATE", "ok")
```

Build ranks, the match score and the net (Part 4, steps 2–4), fill to 5,000 rows, and look at the results block. Answer before you fix anything:

1. How many of your 5,000 hands are **impossible** (the same card twice)? _____
2. Your any-match share vs the exact 49.29%: _____ What does this model say the casino's profit is?
3. In one sentence: what real-world rule did the model forget?

The fix: shuffle a fresh deck for every row and deal from the top. In Google Sheets one formula in B2 fills B2:F2 (clear C2:F2 first — the result *spills* into them), then fill B2 down:

```
=TRANSPOSE(ARRAY_CONSTRAIN(SORT(SEQUENCE(52), RANDARRAY(52), TRUE), 5, 1))
```

Read it inside-out: `SEQUENCE(52)` is the deck 1–52; `SORT(…, RANDARRAY(52), TRUE)` sorts it by 52 random numbers — a shuffle; `ARRAY_CONSTRAIN(…, 5, 1)` keeps the top 5; `TRANSPOSE` lays them across the row. The duplicate count must now be 0 on every recalculation.

4. Build Match Poker

MON JAN 11 · 40 MIN · TAB: POKER

4 One number classifies the hand

HELPER COLUMNS · COUNTIF · LOOKUP

1. **Deal** with the shuffle formula from Part 3.
2. **Ranks** in G2:K2: `=MOD(B2-1,13)+1`, filled right.
3. **Match score** in L2: `=SUMPRODUCT(COUNTIF(G2:K2, G2:K2))`. Each card counts how many cards share its rank (itself included), and the counts are added. Work out a pair by hand: the two paired cards count 2 each, the other three count 1 → 2+2+1+1+1 = 7. Every hand type gets its own score:

NO MATCH	ONE PAIR	TWO PAIR	THREE OF A KIND	FULL HOUSE	FOUR OF A KIND
5	7	9	11	13	17

4. **Hand and Net** (M2, N2): `VLOOKUP` the score in the paytable (columns 2 and 5, exact match, `$`-locked).
5. **Exact EV.** Probability = ways ÷ `COMBIN(52,5)` (that is 2,598,960 possible hands); EV = `SUMPRODUCT` of probabilities and payouts.
6. **Simulated share** of each hand: `COUNTIF` of the score column ÷ hands. Compare with the exact probabilities — this check settles much faster than the money does.
7. Running total, running average, chart — same as dice. Fill to 5,000 hands.

`ways total = 2,598,960` `P(no match) = 50.71%` `P(one pair) = 42.26%`

`exact EV = −0.0384 chips (edge 3.84%)` `0 duplicates`

Expect a wild ride. A 5,000-hand average usually lands between −0.099 and +0.022 — it can even come out positive. Big rare payouts (a +50 four of a kind) make poker the highest-variance game of the three. That is the lesson, not a bug: compare the *frequencies* to check your model, and the long-run average to price the game.

5. Build Blackjack

TUE JAN 12 · 50 MIN · TAB: BLACKJACK

Dice and poker have a formula for the exact answer. Blackjack doesn't — the hitting rules make the possibilities branch too much. **This is the game where simulation is the only practical tool**, which is exactly why casinos used computers to price it.

5 Fixed rules, built column by column

MIXED REFERENCES · NESTED IF

1. **Deal 10 cards** in B2 (player B–F, dealer G–K): the Part 3 formula with `10` in place of `5`. The player uses only as many of B–F as the rules call for; unused cards are never looked at. (Why that is still a fair deal: you'd get the same result dealing in any fixed order from a shuffled deck.)
2. **Values** L2:U2: `=MIN(MOD(B2-1,13)+1, 10)` — Ace = 1 for now, J/Q/K = 10.
3. **Best total after 2, 3, 4, 5 cards** (V2:Y2). An Ace becomes 11 if that does not bust the hand:

```
=IF(AND(COUNTIF($L2:M2,1)>0, SUM($L2:M2)+10<=21), SUM($L2:M2)+10, SUM($L2:M2))
```

Fill it right: `$L2` stays anchored while the end slides to N, O, P — the first 3, 4, then 5 cards.

4. **Player final** (Z2): the first best total that reaches the stand number in the blue cell `$AN$4`, else the 5-card total:

   ```
   =IF(V2>=$AN$4, V2, IF(W2>=$AN$4, W2, IF(X2>=$AN$4, X2, Y2)))
   ```

   A bust stops the hand too, because anything over 21 is also ≥ 17.

5. **Dealer**: repeat steps 3–4 on Q2:U2 with `$AN$5` (AA2:AE2).

6. **Naturals**: AF2 `=V2=21`, AG2 `=AA2=21`.

7. **Result** (AH2), tested in this order: both naturals → Push; player natural → Blackjack; dealer natural → Dealer BJ; player over 21 → Bust; dealer over 21 → Win; then higher total wins, equal is a Push.

8. **Net** (AI2) from the result and the blue cells: Blackjack = bet × natural payout; Win = +bet; Push = 0; anything else = −bet. No typed 2s or 3s.

9. Running total and average, chart, fill to 5,000 rounds. The results block's check cell must say TRUE.

`player natural ≈ 4.6% of rounds` `edge (stand on 17) = 5.57%`

`5,000-round average: usually −0.167 to −0.056 chips`

Strategy experiment. Change only the blue "Player stands on" cell to 12, then 15, then back to 17. Record the simulated edge each time. Can 5,000 rounds tell 15 from 17? (Hint: the 5,000-round range above is about ±2.8 percentage points wide.) On demo day the class pools everyone's rounds to find out.

STANDS ON	12	15	17
Your simulated edge			

6. Convergence, cost, and the Dashboard

TUE JAN 12 · 15 MIN · TAB: DASHBOARD

The *Dashboard* puts the three games side by side. Every yellow cell is a link (`=Dice!L17`) or a formula — never a typed copy of a number.

- **Physical vs simulated vs exact**: one row each, per game. Your 10–15 hand-played rounds will be far from the exact value. That is the point.
- **Dollars per hour** = (chips bet per round) × ($ per chip) × (rounds per hour) × (house edge). Use the pace you timed in Part 1; if you didn't time a game, use the blue assumption cell and say so.
- **Charts**: one convergence chart per game (Parts 2, 4, 5) plus one column chart of house edge by game. Title them with the finding ("Blackjack costs the most per hour"), not the topic ("Blackjack chart").

7. Desmos option (Field Dice only)

OPTIONAL · A SECOND ENGINE FOR THE SAME MODEL

Desmos is good at lists, so it can run the dice game too. Card games need a fresh shuffle per hand, which Desmos lists don't do cleanly — keep poker and blackjack in Sheets. In a new graph at desmos.com/calculator, one line each:

```
N = 10000
a = floor(6 random(N, 1)) + 1
b = floor(6 random(N, 2)) + 1
s = a + b
w = {s = 2: 2, s = 12: 3, 5 ≤ s ≤ 8: -1, 1}
mean(w)
mean(a)
total({a = b: 1, 0}) / N
k = 10          (make it a slider from 1 to 10000)
mean(w[1...k])
```

`random(N, seed)` makes a list of N random numbers between 0 and 1; different seeds give independent dice. **Self-check before you trust it:** `mean(a)` should be near 3.5 and the doubles share near 0.167 (if it says 1, both dice used the same seed). Drag the slider `k` from 10 to 10,000 and watch `mean(w[1...k])` settle toward −0.0278. Change the seeds to re-roll everything.

8. Write-up and the January 14 demo

DUE AT THE END OF CLASS, THU JAN 14

Write-up (on a *Write-up* tab or a linked doc, about one page):

1. **Verdict.** Rank the three games for the Player by house edge, and again by expected loss per hour. Do the rankings agree? Why or why not?
2. **Your night vs the long run.** Compare your physical average to the simulated and exact values for one game. Explain the gap using the size of each sample.
3. **The bug.** What did the with-replacement deal get wrong, what number exposed it, and how would a casino that trusted that model have found out?
4. **Variance.** Which game's running average settled slowest? Use your chart.
5. **Design.** One payout change that gives a game a 1–2% edge. Show it working in the model.
6. **Bridge.** In Cornerstone IV you will simulate stock returns the same way. Name one thing from this lab you will reuse.

Demo (3 minutes per pair, from your Dashboard):

1. **The answer** (30 s): which game is best and worst for the Player, in house edge and in dollars per hour.
2. **Spotlight one game** (60 s): walk one row of its model left to right.
3. **Live change** (60 s): change one blue cell (a payout, the bet, the stand number) and show the EV, edge, and dollars per hour all move. Nothing typed should be left behind.
4. **The chart** (30 s): point to where the running average stopped wandering.

Bring your Blackjack strategy-experiment numbers: the class pools every pair's rounds into one table at the end.

Submit: your Google Sheet link (share: anyone at Cushman with the link can *view*) and this packet.

Rubric — Casino Lab demo

SCORE EACH CRITERION 1–4 · GRADE = AVERAGE

Same four levels as every CS Math cornerstone. Half levels are allowed. Stretch is optional — not attempting it carries no penalty.

CRITERION	4 · ADVANCED	3 · PROFICIENT	2 · DEVELOPING	1 · BEGINNING
Model Correctness Do the three games follow the rules?	All three games match the rules, including naturals, soft Aces and the 5-card stop; exact EVs reproduce −0.0278 and −0.0384; simulated results land in the expected ranges and the frequency checks agree.	All three games run at full length and land in the expected ranges. At most one minor rule slip that does not change the conclusion.	One game is missing or follows the wrong rules, or cards are still drawn with replacement.	Fewer than two games simulate, or results cannot be reproduced.
Formula Honesty Is the arithmetic real?	Every payout, bet and strategy number is a blue parameter that the formulas reference; changing any one re-prices the game end to end. Running totals use the efficient F3 = F2 + E3 pattern.	Formulas do the work and reference the paytables; the Dashboard is all links. A changed payout moves the EV.	Some payouts typed inside formulas (an `IF` with a bare 3), or Dashboard values typed in.	Results typed in; pressing a key to recalculate does not change anything that should change.
Data Structure Could a stranger follow it?	One row per round, labelled helper columns, inputs/model/results visibly separated on every tab; a stranger could audit any row.	Clear headers and the starter's layout kept; inputs and results are easy to find.	Unlabelled columns, stray scratch cells, or rows of different lengths across games.	No discernible structure.
Communication Do the charts, write-up and demo tell the truth?	Convergence charts with the EV line and finding titles; the write-up answers all six prompts with numbers from the model; the live change in the demo works and is explained.	Charts labelled and honest; the write-up states the verdict, the sample-size explanation and the bug; the demo shows a live change.	Charts of running *totals* instead of averages, or conclusions drawn from the physical rounds alone.	No chart or no stated conclusion.
Stretch *(optional)*	Any one, done well: add straights and flushes to Match Poker with a re-priced paytable · sweep the blackjack stand number 12–18 and chart the edge · simulate a bankroll of 20 chips to see how often a player goes broke within an hour · design your own house game with a 1–3% edge and prove it two ways.			

4 Advanced — publishable; someone else could pick this up and use it. **3 Proficient** — works and is honest; the expected standard. **2 Developing** — partly works, or works for the wrong reasons. **1 Beginning** — incomplete, or the numbers cannot be trusted.

Formula card

GOOGLE SHEETS

JOB	FORMULA
One die	`=RANDBETWEEN(1,6)`
Shuffle and deal the top *n* cards across a row	`=TRANSPOSE(ARRAY_CONSTRAIN(SORT(SEQUENCE(52),RANDARRAY(52),TRUE),n,1))`
Rank 1–13 · suit 1–4 · blackjack value	`=MOD(c-1,13)+1` · `=INT((c-1)/13)+1` · `=MIN(MOD(c-1,13)+1,10)`

JOB	FORMULA
Match score (poker) / duplicate test (any 5 cells)	`=SUMPRODUCT(COUNTIF(G2:K2,G2:K2))` — more than 5 means repeats
Look up a payout	`=VLOOKUP(key, $table$, col, FALSE)`
Exact expected value	`=SUMPRODUCT(probabilities, payouts)`
Running total · running average	`F3 = F2 + E3` · `G3 = F3 / A3`
Count / share of an outcome	`=COUNTIF(range, "Win")` · `=COUNTIF(range, ">0") / COUNT(range)`
Recalculate (re-roll everything)	Edit any cell, or press Delete on an empty cell. Sheets re-rolls on every edit.

If the sheet lags: each card row shuffles 52 cards. Build at 200 rows and scale up last. If 5,000 card rows is still too slow on your laptop, use 2,000 and say so in your write-up — an honest smaller sample beats a frozen file. Want to check a game against an independent engine? The LLN simulator includes all three lab games with their exact paytables.